

Supplementary Material: A Controlled Empirical Comparison of Classical and Quantum Kernel SVMs for Breast Cancer Classification

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Provenance: historical release v1.0.0; corrected submission-associated archival release v1.1.0

Release: <https://github.com/SinaQP/SVM-Vs-QSVM/releases/tag/v1.1.0>

Primary Endpoint: Malignant Class F1 Score (pos_label=0)

Supplementary Section S0: Corrected Nested QSVC Selection

For each predefined outer seed, the authoritative QSVC procedure used five-fold stratified inner cross-validation confined to the 455-sample outer-training set. `StandardScaler`, `PCA`, `MinMaxScaler([0,  $\pi$ ])`, `statevectors`, and training/validation Gram matrices were recomputed within every inner fold. The joint grid was $\text{reps} \in \{1, 2, 3\}$ and $C \in \{0.01, 0.1, 1, 10, 100\}$; 4Q additionally searched linear and full entanglement. For 2Q, the topology labels are algebraically equivalent because there is only one qubit pair, so linear is the single canonical representative. Selection maximized mean inner-fold malignant-class F1; ties favored lower repetitions, then linear, then smaller C . The chosen configuration was frozen before full outer-train refitting and one outer-test evaluation.

Across the ten representation/seed selections, one repetition was selected 10/10 times; two and three repetitions were each selected 0/10 times. In 4Q, full was selected 5/5 times and linear 0/5 times. This nested search is the source of the final QSVC models. The separate outer-test ablation remains an exploratory sensitivity analysis and did not choose these configurations.

Table S1: Per-seed corrected nested QSVC selections.

Outer seed	Representation	Reps	Topology	C
42	PCA 2 / 2Q	1	linear	100
123	PCA 2 / 2Q	1	linear	100
456	PCA 2 / 2Q	1	linear	1
789	PCA 2 / 2Q	1	linear	10
2026	PCA 2 / 2Q	1	linear	10
42, 123, 456, 789, 2026	PCA 4 / 4Q	1	full	1

Supplementary Section S1: Extended Sample-Size Scaling Evaluation

Models were trained on nested subsets $N_{\text{train}} \in \{50, 100, 200, 300, 455\}$ with preprocessing pipelines (`StandardScaler`, `PCA`, `MinMaxScaler`) independently refit on each subset and evaluated on fixed outer test partitions ($N_{\text{test}} = 114$). All learning curves use fixed classifier settings: $C = 1$ for every SVC, `gamma='scale'` for RBF, and `reps=1` with full entanglement for QSVC. They are separate from the authoritative nested-selected comparison.

Table S2: Complete Sample-Size Scaling Results across 5 Outer Splits (Mean $\pm$ SD).

N_{train}	Model Architecture	Family	Repr.	Qubits	Test Accuracy	Precision	Recall	Malignant F1	ROC-AUC
50	Linear SVM	Classical	PCA 2	0	0.9333 ± 0.0354	0.8726 ± 0.0699	0.9667 ± 0.0271	0.9157 ± 0.0402	0.9874 ± 0.0087
50	RBF SVM	Classical	PCA 2	0	0.9263 ± 0.0282	0.9006 ± 0.0603	0.9048 ± 0.0532	0.9008 ± 0.0353	0.9839 ± 0.0072
50	QSVC (reps=1, full)	Quantum	PCA 2	2	0.8246 ± 0.0387	0.8242 ± 0.0957	0.6762 ± 0.0643	0.7398 ± 0.0531	0.8850 ± 0.0484
50	Linear SVM	Classical	PCA 4	0	0.9316 ± 0.0353	0.8857 ± 0.0893	0.9476 ± 0.0261	0.9128 ± 0.0395	0.9894 ± 0.0090
50	RBF SVM	Classical	PCA 4	0	0.9421 ± 0.0295	0.9197 ± 0.0607	0.9286 ± 0.0558	0.9222 ± 0.0385	0.9864 ± 0.0060
50	QSVC (reps=1, full)	Quantum	PCA 4	4	0.7263 ± 0.0716	0.6931 ± 0.1444	0.4095 ± 0.2146	0.4994 ± 0.2110	0.7636 ± 0.1082
100	Linear SVM	Classical	PCA 2	0	0.9474 ± 0.0175	0.9311 ± 0.0454	0.9286 ± 0.0412	0.9287 ± 0.0227	0.9884 ± 0.0067
100	RBF SVM	Classical	PCA 2	0	0.9246 ± 0.0237	0.9086 ± 0.0585	0.8905 ± 0.0686	0.8966 ± 0.0317	0.9849 ± 0.0092
100	QSVC (reps=1, full)	Quantum	PCA 2	2	0.8614 ± 0.0190	0.8679 ± 0.0647	0.7429 ± 0.0391	0.7983 ± 0.0207	0.9214 ± 0.0101
100	Linear SVM	Classical	PCA 4	0	0.9421 ± 0.0288	0.9246 ± 0.0675	0.9238 ± 0.0488	0.9222 ± 0.0355	0.9890 ± 0.0074
100	RBF SVM	Classical	PCA 4	0	0.9386 ± 0.0139	0.9256 ± 0.0380	0.9095 ± 0.0639	0.9155 ± 0.0217	0.9884 ± 0.0044
100	QSVC (reps=1, full)	Quantum	PCA 4	4	0.7947 ± 0.0332	0.8059 ± 0.0713	0.6048 ± 0.1722	0.6733 ± 0.0978	0.8628 ± 0.0326
200	Linear SVM	Classical	PCA 2	0	0.9421 ± 0.0159	0.9008 ± 0.0237	0.9476 ± 0.0310	0.9234 ± 0.0213	0.9867 ± 0.0079
200	RBF SVM	Classical	PCA 2	0	0.9316 ± 0.0157	0.9157 ± 0.0451	0.9000 ± 0.0391	0.9066 ± 0.0202	0.9864 ± 0.0045
200	QSVC (reps=1, full)	Quantum	PCA 2	2	0.8877 ± 0.0144	0.8821 ± 0.0521	0.8095 ± 0.0753	0.8406 ± 0.0259	0.9453 ± 0.0115
200	Linear SVM	Classical	PCA 4	0	0.9579 ± 0.0157	0.9432 ± 0.0262	0.9429 ± 0.0213	0.9429 ± 0.0210	0.9910 ± 0.0029
200	RBF SVM	Classical	PCA 4	0	0.9509 ± 0.0159	0.9311 ± 0.0382	0.9381 ± 0.0398	0.9337 ± 0.0216	0.9907 ± 0.0047
200	QSVC (reps=1, full)	Quantum	PCA 4	4	0.8579 ± 0.0287	0.8532 ± 0.0598	0.7476 ± 0.0686	0.7943 ± 0.0440	0.9298 ± 0.0148
300	Linear SVM	Classical	PCA 2	0	0.9439 ± 0.0268	0.9118 ± 0.0589	0.9429 ± 0.0361	0.9258 ± 0.0328	0.9892 ± 0.0038
300	RBF SVM	Classical	PCA 2	0	0.9333 ± 0.0133	0.9242 ± 0.0403	0.8952 ± 0.0494	0.9080 ± 0.0191	0.9864 ± 0.0047
300	QSVC (reps=1, full)	Quantum	PCA 2	2	0.9158 ± 0.0260	0.9193 ± 0.0448	0.8476 ± 0.0598	0.8807 ± 0.0388	0.9651 ± 0.0130
300	Linear SVM	Classical	PCA 4	0	0.9596 ± 0.0078	0.9441 ± 0.0240	0.9476 ± 0.0310	0.9453 ± 0.0112	0.9943 ± 0.0019
300	RBF SVM	Classical	PCA 4	0	0.9544 ± 0.0144	0.9493 ± 0.0444	0.9286 ± 0.0376	0.9377 ± 0.0188	0.9936 ± 0.0037
300	QSVC (reps=1, full)	Quantum	PCA 4	4	0.8877 ± 0.0259	0.8725 ± 0.0539	0.8190 ± 0.0598	0.8429 ± 0.0351	0.9489 ± 0.0156
455	Linear SVM	Classical	PCA 2	0	0.9474 ± 0.0152	0.9152 ± 0.0444	0.9476 ± 0.0261	0.9302 ± 0.0182	0.9897 ± 0.0026
455	RBF SVM	Classical	PCA 2	0	0.9439 ± 0.0159	0.9339 ± 0.0411	0.9143 ± 0.0271	0.9233 ± 0.0202	0.9864 ± 0.0040
455	QSVC (reps=1, full)	Quantum	PCA 2	2	0.9123 ± 0.0224	0.9313 ± 0.0269	0.8238 ± 0.0706	0.8726 ± 0.0376	0.9697 ± 0.0141
455	Linear SVM	Classical	PCA 4	0	0.9632 ± 0.0157	0.9493 ± 0.0344	0.9524 ± 0.0376	0.9501 ± 0.0211	0.9952 ± 0.0030
455	RBF SVM	Classical	PCA 4	0	0.9579 ± 0.0200	0.9448 ± 0.0436	0.9429 ± 0.0361	0.9430 ± 0.0259	0.9944 ± 0.0035
455	QSVC (reps=1, full)	Quantum	PCA 4	4	0.9053 ± 0.0423	0.8748 ± 0.0767	0.8762 ± 0.0832	0.8721 ± 0.0556	0.9581 ± 0.0227

Supplementary Section S2: Computational Complexity and Execution Profiling

The reported `total_runtime` includes preprocessing, classifier fitting, and prediction; QSVC additionally includes exact statevector generation and both training and test Gram products. Inner-search and kernel-diagnostic time are excluded.

Table S3: Execution Time, Computational Complexity, and Storage Footprint.

Pipeline Stage / Architecture	Platform	Qubits	Mean Runtime (s)	Rel. Cost	Time Complexity	Memory Complexity
Linear SVM (PCA 2)	CPU LibSVM	0	0.0048 ± 0.0007	$1.0\times$ (Ref)	$\mathcal{O}(dN_{tr}^2) - \mathcal{O}(dN_{tr}^3)$, data dependent	Data/cache dependent
RBF SVM (PCA 2)	CPU LibSVM	0	0.0072 ± 0.0024	$1.5\times$	Same SVC bounds; pairwise RBF work	Data/cache dependent
Linear SVM (PCA 4)	CPU LibSVM	0	0.0083 ± 0.0061	$1.7\times$	$\mathcal{O}(dN_{tr}^2) - \mathcal{O}(dN_{tr}^3)$, data dependent	Data/cache dependent
RBF SVM (PCA 4)	CPU LibSVM	0	0.0071 ± 0.0019	$1.5\times$	Same SVC bounds; pairwise RBF work	Data/cache dependent
QSVC (2Q Statevector)	Vectorized CPU	2	0.2181 ± 0.0096	$\sim 45\times$	$\mathcal{O}((N_{tr}^2 + N_{te}N_{tr})2^q)$ plus SVC	Train+test kernels: 2,071,160 B
QSVC (4Q Statevector)	Vectorized CPU	4	0.5846 ± 0.0257	$\sim 71\times$	Same form with larger q plus SVC	Train+test kernels: 2,071,160 B
<i>Historical 2Q ComputeUncompute</i>	Local circuit-pair sampler	2	120.5 ± 15.2	$\sim 25,000\times$	$\mathcal{O}(N_{tr}^2 + N_{te}N_{tr})$ pairs	Train+test Gram arrays
<i>Historical 4Q ComputeUncompute</i>	Local circuit-pair sampler	4	455.0 ± 40.0	$\sim 55,000\times$	$\mathcal{O}(N_{tr}^2 + N_{te}N_{tr})$ pairs	Train+test Gram arrays

The exact statevector and historical `ComputeUncompute` measurements are separate CPU-simulation regimes. Neither is a physical-QPU timing, and no fixed hardware shot budget is inferred.

Supplementary Section S3: Supplementary Diagnostic Figures

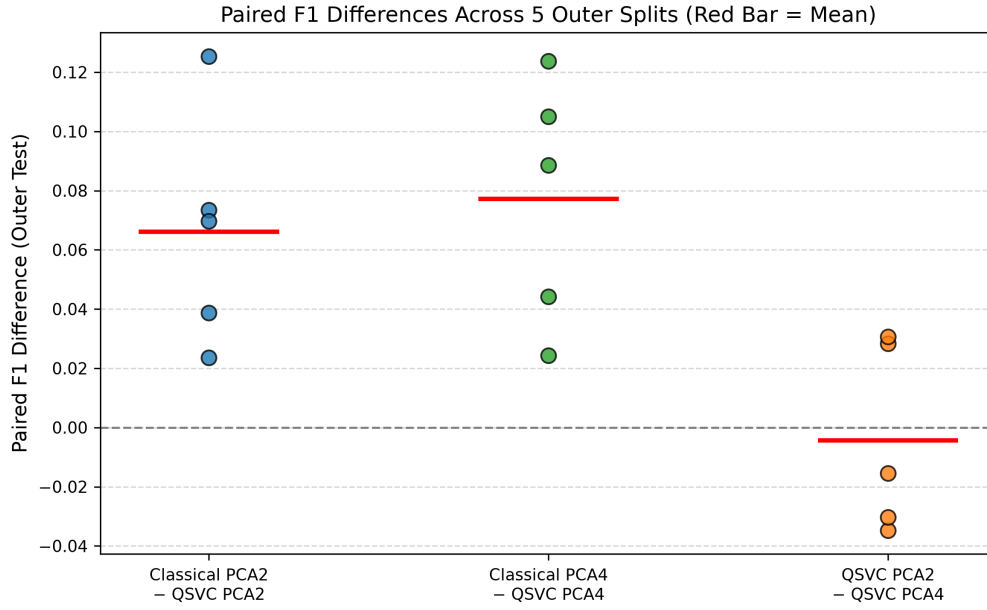


Figure S1: **Paired Split-Level F1 Differences.** Distribution of paired differences $\Delta = F1_{\text{Classical}} - F1_{\text{QSVC}}$ across five outer test splits. Classical models achieve higher malignant F1 scores on 5/5 splits in both PCA 2 and PCA 4 representations.

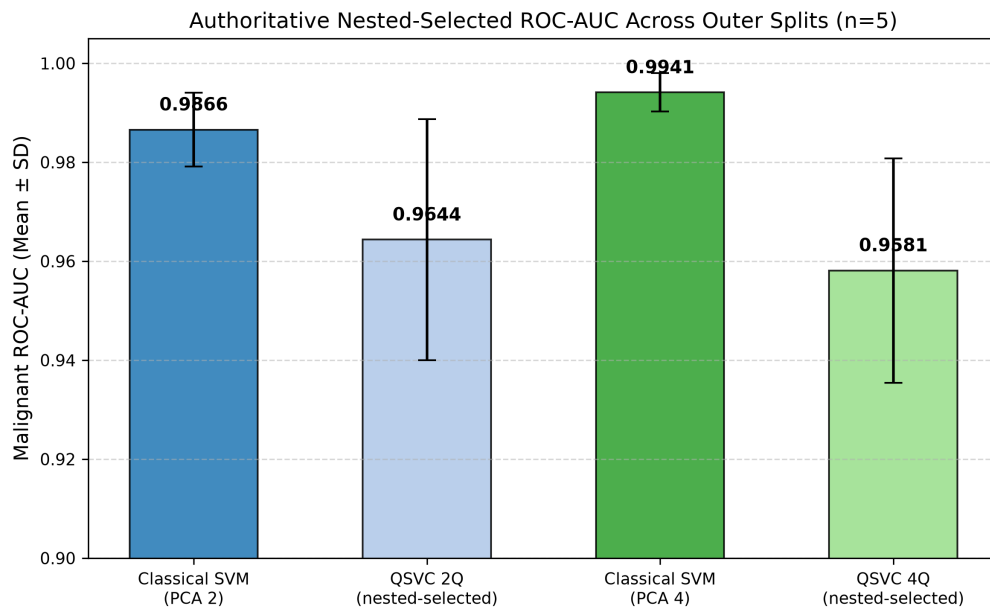


Figure S2: **Outer-Test ROC-AUC Comparison.** Mean receiver operating characteristic area under curve across five outer splits comparing classical and quantum models.